Youngjin Moon

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Summary

- **Expertise in AI/ML Acceleration ASIC Design**
  - Digital Deep Learning Accelerator
  - Hardware-Software Co-design (Skilled in sparsity optimization and model compression)
- **End-to-end IC Implementation & Verification**
  - Multiple Tape-outs including Video AI accelerator and Large Language Model accelerator
  - Full stack digital (RTL, Synthesis, PnR) chip design experiences

Education

- **Ph.D Candidate in AI Semiconductor, KAIST** — *Mar. 2026 -* — Daejeon, Republic of Korea
  - Advisor: Hoi-Jun Yoo, KAIST (IEEE Fellow since 2008)
- **M.S in AI Semiconductor, KAIST** — *Mar. 2024 - Feb. 2026* — Daejeon, Republic of Korea
  - Advisor: Hoi-Jun Yoo, KAIST (IEEE Fellow since 2008)
  - Thesis: An Energy-Efficient Real-Time Mamba-Based Video Understanding Accelerator for Edge Devices
- **B.S in Mechanical Engineering & Electrical Engineering, SKKU** — *Mar. 2017 - Feb. 2024* — Suwon, Republic of Korea
  - GPA: 4.03/4.50
  - Major1(2) GPA: 4.01(4.05)/4.50

Work Experience

- **Samsung Electronics** — *Aug. 2022 - Dec. 2022*
  Long-term Internship in Modem Development Team, System LSI — Suwon, Republic of Korea
  - Analyze High Speed Digital Interface Architecture
  - Design Sub-module of High Speed Digital Interface
- **Republic of Korea Army** — *Sep. 2018 - Apr. 2020*
  Mandatory Military Service — Cheonan, Republic of Korea
  - Trained, disciplined, and instructed soldiers in various military skills
  - Gained leadership and operational experience as a Duty Officer

Skills

- **ASIC Design:** Digital Processor & Deep Learning Accelerator Design, Hardware-Software Co-design
- **Programming Languages:** Verilog HDL, Python, C/C++
- **EDA Tools:** Synopsis Design Compiler, IC Compiler II, PrimeTime, Cadence Virtuoso
- **Language:** English (fluent), Korean (native)

Representative Publications

C=Conference, J=Journal, P=Patent, S=In Submission, T=Thesis

[C.1] **[S.VLSI 2026]** **Youngjin Moon**, Sangwoo Ha, Junha Ryu, Jungjun Oh, Jingu Lee, Wooyoung Jo, Minsung Kim, and Hoi-Jun Yoo. *"NOVA: A 5.3mJ/frame Fully Integrated Neural Video SoC with Content-Adaptive Architecture for 4K Real-time Encoding/Decoding and Processing"*, 2026 IEEE Symposium on VLSI Technology and Circuits (S.VLSI)

[C.2] **[ISCAS 2025]** **Youngjin Moon**, Sangwoo Ha, Soyeon Kim, Junha Ryu, Hoi-Jun Yoo, and Donghyeon Han. *"A 2.67 mJ/frame Video Mamba Accelerator with Importance-aware Redundancy Elimination and SSM Computing Reformulation"*, 2025 IEEE International Symposium on Circuits and Systems (ISCAS)

[C.3] **[ASSCC 2026]** Jungwan Lee, Seongyon Hong, **Youngjin Moon**, Jingu Lee, Junhyuk Lee, Yuseon Choi, Jungjun Oh, and Hoi-Jun Yoo. *"VectorFLASH: a 2.8-8.7 Tokens/S Resource-Efficient Long-Context LLM Accelerator with Dual Vector Quantization on Edge FPGA "*, 2026 IEEE Asian Solid-State Circuits Conference (ASSCC)

[C.4] **[MICRO 2026]** Sangwoo Ha, Hyunwoo Seo, Yurim Jo, **Youngjin Moon**, and Hoi-Jun Yoo. *"EdgeXpert: An Edge Device for Memory-Efficient LLM Inference with Mixture-of-Experts and Speculative Decoding"*, 2026 IEEE/ACM International Symposium on Microarchitecture (MICRO)

[C.5] **[COOLCHIPS 2026]** Sangwoo Ha, Jingu Lee, **Youngjin Moon**, Sunjoo Whang, Wooyoung Jo, Gwangtae Park, Soyeon Um, Junha Ryu, Yurim Jo, Jongjun Park, and Hoi-Jun Yoo. *"An Energy-Efficient LLM Processor for Mobile Devices with Pipelined NPU-CIM"*, 2026 IEEE Symposium on Low-Power and High-Speed Chips (COOL Chips)

[C.6] **[ISSCC 2026]** Sangwoo Ha, Jingu Lee, **Youngjin Moon**, Sunjoo Whang, Wooyoung Jo, Gwangtae Park, Soyeon Um, Junha Ryu, Yurim Jo, and Hoi-Jun Yoo. *"SMoLPU: A 122.1 μJ/token Sparse MoE-based Speculative Decoding LLM Processing Unit with Adaptive-Offload NPU-CIM Core"*, 2026 IEEE International Conference on Solid-State Circuits (ISSCC)

[C.7] **[S.VLSI 2025]** Soyeon Kim, Hankyul Kwon, Jingu Lee, **Youngjin Moon**, Hongseok Lee, Junha Ryu, Zhamaliddin Kalzhan, Sangyeob Kim, Wooyoung Jo, and Hoi-Jun Yoo. *"NuVPU: A 4.8 9.6 mJ/frame Progressive NTT-based Unified Video Processor for Stable Video Streaming and Processing with Neural Video Codec"*, 2025 IEEE Symposium on VLSI Technology and Circuits (S.VLSI)

[J.1] **[TCSVT 2026]** **Youngjin Moon**, Sangwoo Ha, Junha Ryu, Soyeon Kim, Donghyeon Han, and Hoi-Jun Yoo. *"An Energy-Efficient State-Space Model Accelerator for Real-Time Video Understanding via Redundancy-Conscious Design"*, IEEE Transactions on Circuits and Systems for Video Technology, June. 2026

[J.2] **[TCAS-II 2025]** Sangmyoung Lee, Seryeong Kim, Jongjun Park, **Youngjin Moon**, Minsung Kim, Junha Ryu, Hoi-Jun Yoo, Donghyeon Han. *"A 227.1 TOPS/W High Energy-efficiency PNN-based 3D Object Recognition Processor with Spiking Neural Network for Edge Device"*, IEEE Transactions on Circuits and Systems II, Jul. 2025

HONORS AND AWARDS

- **Academic Excellent Scholarship**
  Sungkyunkwan University, 2017-1st semester GPA 4.5/4.5 — Sep. 2017
- **Global Student Success Creative Scholarship**
  Sungkyunkwan University — Sep. 2021
- **S-Hero Gold Prize**
  Ministry of Science and ICT of Korea, Field Customized Science and Engineering Project — Sep. 2021
- **Intelligent Connect Leading University Competition Grand Prize**
  Sungkyunkwan University, Creative Research Presentation Competition — Oct. 2021
- **X-Corps Festival Grand Prize (Award of Ministry of Science and ICT)**
  Ministry of Science and ICT of Korea, Undergraduate Research Project Contest — Nov. 2021
- **SAMSUNG Dream Class Mentor Scholarship**
  SAMSUNG, Mentoring middle school students — Nov. 2022
- **IEEE ISCAS Student Travel Grant Award**
  IEEE Circuits and Systems Society (CASS) — Apr. 2025
- **IITP President's Award**
  Institute for Information & communication Technology Planning & evaluation(IITP) of Korea, AI Semiconductor Contest — Jun. 2025

REFERENCES

1. **Hoi-jun Yoo**
   IEEE Fellow, Professor, KAIST
   Email: hjyoo@kaist.ac.kr
   Thesis Advisor